

Diagnostic Accuracy of Artificial Intelligence- Enabled Screening for Oral Cancer: A Systematic Review and Meta-analysis

Dahy Sulaiman

Indian Institute of Science Bangalore

Tejashree Subramanya

Indian Institute of Science Bangalore

Anushka Amble

Indian Institute of Science Bangalore

Sumirtha Gandhi

sumirtha@iisc.ac.in

Indian Institute of Science Bangalore

Debnath Pal

Indian Institute of Science Bangalore

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Abstract

Background

Oral cancer is a global health challenge, with delayed detection leading to poor survival. Artificial intelligence (AI)-based tools show promise for early detection, but evidence remains fragmented and heterogeneous. This systematic review and meta-analysis evaluated the diagnostic performance and clinical utility of AI-enabled approaches for oral cancer screening.

Methods

PubMed, Scopus, Web of Science, Cochrane Library, and Google Scholar were searched through August 2024. Study quality was assessed using the QUADAS-2 tool.

Results

Twenty-seven studies were included in the review, with nine contributing to the meta-analysis. The pooled sensitivity for the reviewed AI-enabled screening systems was 0.91 (95% CI: 0.81–0.91), the pooled specificity was 0.91 (95% CI: 0.86–0.94), and the area under the summary receiver operating characteristic curve was 0.938, indicating excellent diagnostic accuracy.

Conclusion

The findings of this study demonstrate the high diagnostic accuracy of AI-based technologies for oral cancer screening and underscore their potential for integration into clinical practice to enhance early detection and improve patient outcomes.

BACKGROUND

Oral cancer represents a significant global health challenge, characterised by its aggressive nature and often late-stage diagnosis. This, in turn, contributes to poor prognosis and reduced survival rates.(1) The predominant type, oral squamous cell carcinoma (OSCC), accounts for over 90% of cases.(2) According to GLOBOCAN 2022 data, there are approximately 389,846 new cases annually, with 188,438 deaths attributed to this disease (<https://gco.iarc.fr/to>). The global burden of oral cancer is especially pronounced in South Asia, where it constitutes a major public health concern, driven by high-risk behaviours, such as tobacco use, betel quid chewing, and human papillomavirus infection.(3) Despite advancements in cancer prevention and therapeutic strategies, the 5-year survival rate remains below 50% due to late diagnoses and high rates of recurrence or metastasis, emphasising the urgent need for improved early detection methods.(4)

Conventional oral cancer screening methods, which mainly rely on visual oral examinations, face challenges in reaching rural and remote populations, and telemedicine services have demonstrated potential in addressing these accessibility issues. Conventional methods are also hindered by their subjective nature and low sensitivity, which can lead to delayed diagnosis and adversely affect patient outcomes.(5) These issues can be overcome by employing advanced technologies such as artificial intelligence (AI)-driven screening methods, which hold significant potential, especially in settings with limited resources and healthcare access.(6) The integration of AI into healthcare, particularly in diagnostics, holds tremendous promise for revolutionising disease screening. AI algorithms, including machine learning and deep learning models, have demonstrated remarkable capabilities in analysing complex datasets, identifying subtle patterns, and assisting clinicians in making more accurate and timely diagnoses.(7) AI-enabled tools can analyse medical imaging data and other test data with precision, detecting early signs of disease that may be missed by human examination.(8) Furthermore, AI algorithms can be trained to recognise high-risk features in the population, predict disease progression, and personalise treatment strategies, thereby improving patient care and outcomes.(9)

Considering the limitations of traditional screening methods, the escalating burden of oral cancer, and the transformative potential of AI, there is a compelling need for a comprehensive evaluation of the diagnostic accuracy of AI-enabled tools in oral cancer screening. Hence, we conducted this systematic review and meta-analysis with the aim of assessing the diagnostic accuracy of AI-enabled screening tools in identifying oral cancer and pre-cancerous lesions. By synthesising sensitivity and specificity data from existing literature, this review seeks to establish the clinical utility of AI in early cancer detection and explore its role as a viable alternative or complement to traditional screening methods, particularly in settings with limited healthcare resources. These findings will also provide evidence supporting economic modelling and informing healthcare policy, reinforcing AI's potential to improve patient outcomes.

METHODS

The protocol of this systematic review was registered with PROSPERO (CRD42024577773).

Search strategy

Terms related to “oral cancer”, “artificial intelligence”, “screening”, and “diagnostic accuracy” were searched in PubMed, Scopus, Web of Science, Cochrane Library, and Google Scholar. Studies published up to August 2024 were included in the review, without any language restrictions. Search results from electronic databases were exported into the Rayyan software.(10)

Selection criteria

The review included studies fulfilling the following criteria: (1) individuals screened for oral cancer or oral pre-malignant conditions, (2) screening conducted with artificial intelligence enabled screening tools for oral cancer (3) the diagnostic accuracy of the test was compared with conventional screening methods (without AI), an evaluation by a specialist/dentist, histopathological examination, or biopsy as the reference standard, (4) sufficient data were available to calculate sensitivity, specificity, PPV, and NPV, and (5) studies published in English language or summary in English. Exclusion criteria were: (1) studies on cancers other than oral or oral pre-malignant conditions, (2) screening tools that do not use AI, (3) studies on intervention with AI tools that can be used for diagnosis, not for screening, and (4) Conference proceedings, reviews, and case studies.

Study selection and data extraction

After duplicates were removed from the initial search results, two reviewers independently screened the articles according to the selection criteria. Any disagreements were resolved by the third author. The authors of the studies included were contacted to request additional data that was not available in the reports. Two reviewers used a pretested spreadsheet to independently extract data from the reports included. Information on the publication, country, details of index tests, comparator, the disease being studied, and outcome estimates, such as TP, TN, FP, and FN, was collected.

Study quality assessment

The methodological quality of the included studies was evaluated using the Quality Assessment of Diagnostic Accuracy Studies 2 QUADAS-2 tool. Risk of bias was judged as high, low, or unclear across four key domains: (1) patient selection, (2) index test, (3) reference standard, and (4) flow and timing. All domains were evaluated for risk of bias, and the first three domains were additionally examined for applicability concerns.⁽¹¹⁾

Statistical analysis

Indicators of diagnostic accuracy, i.e., the number of TPs, TNs, FPs, and FNs for each test in each study, were used to calculate summary estimates of sensitivity and specificity with 95% confidence intervals. Where required, the number of TPs, TNs, FPs, and FNs was calculated from the reported confusion matrices. For the purposes of statistical analysis, all pre-cancerous (i.e., oral pre-malignant diseases, OPMD) and cancerous lesions were considered positives, and normal, benign, and other variational presentations were considered negatives. Forest plots were used to graphically display the point estimate, as well as the confidence intervals for sensitivity and specificity. A summary receiver operating characteristic curve was also plotted for the joint distribution. All statistical analyses were performed using STATA version 18 (Stata Corp., College Station, TX, 135 USA) and R (version 4.3.3; R Foundation for Statistical Computing, Vienna, Austria).

RESULTS

A total of 620 articles were identified as potentially relevant for inclusion in the review. After the removal of duplicate records, 386 articles were retained for initial screening. Of these, 62 met the criteria for full-text review. Following assessment of the full texts, 35 articles were excluded, resulting in a final inclusion of 27 studies in the systematic review (Figure 1). The primary reason for excluding articles at the full-text stage was that the intervention tool assessed was designed for diagnosis rather than screening (n = 13), followed by discrepancies between the study objectives and the review question (n = 12). Nine articles were included in the quantitative synthesis (meta-analysis). The Preferred Reporting Standards for Systematic Reviews and Meta-analyses (PRISMA) flow diagram summarising the study selection process is presented in Figure 1.

Overview of included studies

The articles reviewed in this systematic review were published in the period from 2013 – 2024. All 27 articles were cross-sectional studies. Of these, 12 studies were based on datasets sourced from India, three each from China and Thailand, and one each from Australia, the USA, Egypt, Germany, Israel, and Japan; two studies leveraged combined datasets from Australia-USA and Jordan-Greece, while some studies leveraged datasets available online (Figure 2). The main characteristics of studies are shown in Table 1.

Quality assessment

The methodological quality and risk of bias of the included diagnostic accuracy studies were evaluated using the Quality Assessment of Diagnostic Accuracy Studies-2 (QUADAS-2) tool, which systematically assesses four key domains: patient selection, index test, reference standard, and flow and timing.⁽¹²⁾ A low risk of bias across all domains was observed in 76% (n = 22) of the studies. For the ‘patient selection’ domain, the risk of bias could not be determined in 18% (n = 5) of studies due to insufficiently clear reporting on potential patient exclusions and the sampling method employed. All other domains demonstrated a consistently low risk of bias and minimal applicability concerns across all studies.

A summary of the risk of bias assessment for each study is presented in Figure 3. Overall, the quality assessment indicated that while most studies had a low risk of bias across domains, some methodological limitations—particularly regarding reporting of patient selection and patient flow—should be considered while interpreting the results of this review.

Qualitative synthesis

In this review, we observed that the AI tools developed in the 27 articles were for either classification (binary/multi-class) or detection of oral lesions. The binary classes considered were of two types: oral pre-malignant lesion/oral cancer vs. normal/benign/others and oral cancer vs. normal/benign/others. The most used input modality for screening was clinical images. Many of the studies that used image datasets reported only information pertaining to the

datasets used for model development, without providing patient-level characteristics corresponding to the images. Among the included studies, 12 studies reported patient details, such as the number of patients enrolled for the study, and patient-wise lesion distribution.(13–24) The rest of the studies only reported the size of the image dataset used for AI model development, which ranged from 60 to 7970 images. The study population consisted of individuals with a history of tobacco use and those with any type of oral lesion in four of the studies(16,17,25,26), while the rest studies sourced screening tests data from hospitals and colleges, publicly available datasets, or searches on the internet.

AI models

A total of 66 AI models, trained and tested across the 27 included studies, were considered in this review. Four studies(15,24,26,27) utilised machine learning (ML) algorithms, such as Light Gradient Boosting Machine (LightGBM), L2-regularised logistic regression and Support Vector Machine for the screening task. Most models developed for oral cancer screening were based on deep learning (DL) models. For classification tasks, DL models based on transformer architecture: Vision Transformer (ViT), Swin Transformer, and DL models based on convolutional neural network (CNN) architecture: High-Resolution representation learning network (HRNet), Residual Network (ResNet), Densely Connected Convolutional Network (DenseNet), MobileNet, VGG16/19, Single Shot Multibox Detector, EfficientNet-B0, Radial Basis Function Networks were leveraged. CNN models such as You Only Look Once, Faster R-CNN, and Mask R-CNN were used for lesion segmentation tasks. One study leveraged hybrid architectures, such as the Mask R-CNN model with the small variant of the Swin Transformer as the backbone, for multi-class classification and delineation of oral lesions.(28)

Ensembling and cascading methods

Rabinovici-Cohen et al., Nanditha et al., and Shamim et al. used ensemble methods with different strategies. In the study by Rabinovici-Cohen et al., the two best patient models were ensembled based on the patient's score, which was determined as the mean of all the image scores associated with that patient. Nanditha et al. ensembled two CNN architectures, and Shamim et al. formed an AI + Physician ensemble by placing an additional physician in the loop after model classification to reduce the number of misclassifications. These studies found that ensemble methods performed better than individual models. (14,29,30)

Fu et al. developed a cascaded CNN network — the Single Shot MultiBox Detector with VGG16 — as a backbone network for lesion detection (by generating a bounding box that located the suspected lesion), followed by a DenseNet-based classifier to detect OSCC from photographic images (the cropped lesion area was classified into OSCC or non-OSCC lesions), to form a model system capable of both segmentation and classification.(31)

Datasets

While several studies used data obtained through clinical studies for the development of AI models, some studies collected data from Google search results(29) and other internet-based searches(21) to gather data that was later reviewed by specialists. Sampath et al. leveraged a publicly available online dataset (Oral Cancer [Lips and Tongue] images, Kaggle) that noted uncertainty about its reliability.(32) All studies primarily used imaging of the oral cavity as the input modality, except Banavar et al. (24,26) and Shree et al.(15), where saliva samples were collected to obtain metatranscriptomic signatures and Fourier transform infrared spectroscopy, respectively, which were later used as inputs to the models. Among the imaging-based studies, white light-based imaging (WLI) was the most widely used imaging modality, and four studies(16,18,19,22) leveraged autofluorescence imaging (AFI) in addition to WLI. A dual-view image capture using a whole-cavity imaging module (smartphone-based WLI) and an intraoral probe module (near field AFI) was conducted. A new 3-channel image was created by combining red and green channels from WLI and the normalised ratio of red and green channels from AFI, which was then fed to the model. Similarly, Yang et al.(17) utilised polarised WLI and AFI to capture pairs of oral cavity images for risk analysis.

Pre-processing techniques

To enhance model performance, some studies explored supplementary pre-processing methods before image input, beyond the standard model-mandated steps (e.g., image resizing and normalisation). Nanditha et al. reported that image pre-processing techniques based on frequency and spatial domain helped in performance gains.(30) Dutta et al. used Gaussian filtering to reduce image noise.(23) Goswami et al. explored five distinct colour spaces, Red-Green-Blue, Hue-Saturation-Value, YCbCr, Hue-Saturation-Lightness and XYZ colour spaces for the extraction of colour and texture features.(27) Mira et al. and Lin et al. reported that centre image positioning and resampling techniques during image capture led to better results; Mira et al. also used centre cropping to enhance diagnostic features in the images.(13,33)

Comparator

In classification studies, histopathological reports were used as the ground truth.(13–15,20,21,24,26,31,33–36,36,37) Other studies used telediagnosis by oral specialists(16,25), assessments by an on-site specialist(17,22), annotations by oral medicine specialists(27,38), clinical diagnoses by oral oncology specialists(18,19), annotation by a physician(29), and assessment by an ear-nose-throat specialist.(32) Nanditha et al. sourced labelled image data from medical colleges and hospitals and did not specify the comparator used.(30) In detection studies, the experts annotated the lesion area in addition to providing diagnostic labels, which were further used for developing classification and segmentation models. The largest intersection of annotated areas or a review by another expert served as the ground truth for segmentation.(28,34–36,39)

Performance metrics

The metrics of performance were reported in terms of accuracy, sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), F1 score, precision, and area under the receiver operator characteristic curve. In segmentation studies, additional metrics such as Intersection over Union (IoU: AP50, AP50:95) were reported. A few studies also reported true positive (TP), true negative (TN), false positive (FP) and false negative (FN) values, which were considered for the quantitative synthesis.

Accuracy was reported as one of the performance metrics for 32 models, with values varying from 36% to 99.9%, and a mean accuracy of 80.3%. Sensitivity was reported for all the models, and the values ranged from 0% to 100% with a mean value of 77.7%. Specificity was reported for 55 models, with values ranging from 9% to 100% and a mean of 78.4%. Fifteen models reported PPV and NPV, with values ranging from 22% to 87% and 71% to 100%, and mean values of 60.4% and 83.8%, respectively.

Model evaluation studies

Some studies performed evaluations of the AI models, experimenting with various AI model architectures, for classification and detection tasks. These studies reported metrics associated with each model and highlighted the model with the best performance.(13,21,29,33,35–38) While Shamim et al. (13,21,29,33,35–38) and Sharma et al.(13–15,20,21,24,26,31,33–36,36,37) reported VGG19 to have the best performance in comparison with networks like ResNet50 and Inceptionv3, Mira et al.(13,21,29,33,35–38), Talwar et al.(27,38), Lin et al.(13,21,29,33,35–38), and Jubair et al. ¹⁸ reported that models like HRNet-W18, DenseNet201, and EfficientNet-B0 had better or similar performance in comparison with VGG19, highlighting that small networks can provide similar performance as that of the large models, with an additional feasibility of low-computing, low-memory, embedded applications.

Risk-based stratification of the target population

A few papers reported model evaluation studies on different sub-groups within the study population. Yang et al. developed a screening algorithm for tobacco users and non-user groups, with a superior agreement with specialist screening outcomes in tobacco smokers than in non-smokers, highlighting that tobacco use needs to be considered carefully in the development of any imaging-based screening algorithm.(17) Banavar et al. developed ML-based tools for 1. Screening for OC/OPMD within the high-risk population, 2. Screening for OC is only done within the average-risk population. The ML classifiers were evaluated in four cohorts, Cohort A, B, C and D, based on risk identified by age and tobacco usage. The performance of the models in screening, for OC, OPMD in cohorts A, B and for OC in cohorts C, D indicate similar accuracies, with the models having better sensitivity in high-risk cohorts and having better specificity in average-risk cohorts.(24) The findings from these studies indicate that evaluating AI tools in different populations provides more utility.

Inclusion of metadata for AI inference

While most studies did not consider the metadata associated with images for classification tasks, one study developed alternate models incorporating late fusion of metadata, encompassing lesion type and lesion location, into ResNet architectures.¹¹ There were instances when the patient-based, metadata-inclusive models performed better than those that did not consider the metadata. This highlights the potential of models to pick up correlations between input data and the associated metadata, bringing in performance improvements.

Interpretability of AI models

Talwar et al.³² and Warin et al.(38,39) investigated the interpretability of the developed AI models by generating heatmaps via GradCAM-based methods and compared them with the specialist-annotated lesion areas. They demonstrated examples where the generated heatmaps aligned with the lesion areas annotated by oral specialists, adding reliability to the model's inferences.

External validation

While most studies were restricted to testing the developed AI models on internal validation/test datasets, which likely had characteristics similar to those of the training data, Talwar et al.(31,38) and Fu et al.(31,38) conducted model validation studies on external datasets. Talwar et al. reported reduced performance on the external validation dataset, and Fu et al. reported a slight reduction in performance on one of the external validation datasets, and an improvement in performance in the other set, also called the clinical validation set, which was used to conduct comparative studies between AI and human readers. It was also reported that the AI algorithm achieved performance comparable to that of the average oral cancer expert and significantly better than an average medical and non-medical student. Further, Birur et al. conducted a field validation study where data collected from the field was evaluated by field health workers who were involved in data collection, on-site oral specialists, remote specialists, a MobileNet-based model on the phone and a VGG19 model with Bayesian prediction on the cloud. Intergroup comparative results were reported in the paper, suggesting good performance of AI models as compared to teleradiology.(16) Similarly, Ye et al. made a compared YOLOX-based AI models developed for lesion detection with groups of clinicians with varying levels of expertise. They found that the performance for the AI surpassed that of the senior experts. A multi-centric AI-assisted diagnosis evaluation revealed that the dental, general, and community hospital groups aided by the AI model significantly improved performance.(34)

Other relevant findings

Warin et al. conducted three studies to classify and detect OPMD and OSCC. They found that although models displayed reasonably good sensitivities in detecting OSCC lesions, the values dropped when detecting OPMD lesions across all models. In contrast, no class-wise difference was observed in classification performance, with models having similar performance metrics between the two classes.(35,36,39)

Song et al. and Talwar et al. leveraged similar datasets for model development. They independently developed AI models based on ViT, Swin transformer, DenseNet-201, and other model architectures for the binary classification of oral cavity images into 'Suspicious' and 'Non-suspicious' categories, with the 'Suspicious' category containing oral potentially malignant and malignant lesions. Both studies reported good performance in terms of sensitivity and specificity, with transformer-based models providing slight performance improvements over CNN models.(25,38)

Kouketsu et al. developed models for binary classifications of OSCC vs. Normal and OSCC + Leukoplakia vs. Normal. In the latter classification, sensitivity dropped as the false negatives increased, possibly due to the misclassification of leukoplakia cases. At the same time, the specificity was maintained at the

same level, indicating that models might get confused while classifying borderline cases, highlighting the need for further studies on the misclassified images to find potential features leading to misclassification.(20)

Quantitative synthesis

The meta-analysis included 9 studies with 16 models for which TP, TN, FP, and FN values were reported. The pooled sensitivity was 0.91 (95% CI: [0.81 – 0.91], $\text{Tau}^2 = 2.44$; $I^2 = 71.87\%$; Figure 4), while the pooled specificity was 0.91 (95% CI: [0.86 – 0.94], $\text{Tau}^2 = 0.74$; $I^2 = 51.76\%$, Figure 4). The area under the summary receiver operating characteristic curve was 0.938, demonstrating excellent diagnostic accuracy (Figure 5).

Publication bias

Publication bias was assessed using Deeks' funnel plot asymmetry test. The test showed significant asymmetry ($p < 0.05$), indicating a high likelihood of publication bias among the included diagnostic accuracy studies. This suggests that smaller studies with non-significant or unfavourable results may be underrepresented, potentially inflating the pooled diagnostic odds ratio. Therefore, the summary estimates should be interpreted with caution.

DISCUSSION

This systematic review and meta-analysis, which synthesised evidence across 27 studies and 66 diagnostic models for AI-enabled oral cancer screening, demonstrated strong overall diagnostic performance. Among the studies included in the meta-analysis, sensitivity and specificity were generally high, indicating that the test was effective at correctly identifying both true cases and non-cases. While variability across studies was observed, the findings highlight the potential of this approach as a reliable screening tool.

Kavyashree et al. conducted a systematic review of 73 studies on AI techniques for oral cancer detection. Different types of input modalities, including histopathological images, white light images, genetic data, X-ray images, CT images, and others, were included in the review. However, only studies with reported accuracies greater than 80% were included, which may have introduced a bias to their findings.(40) On the other hand, Li et al. conducted a systematic review of 17 studies on AI-assisted clinical imaging for the detection of OPMD and OC, focusing only on studies that included both OPMD and OC with imaging as the input modality.(41)

Our systematic review exclusively focuses on AI tools for oral cancer screening; therefore, studies focused on tools for diagnosing oral cancer were excluded. No conditions were imposed on the input modality or the reported performance metrics. Studies based on modalities such as imaging and saliva data, as well as all models reported in the studies, irrespective of their performance and classification criteria (identifying either OPMD, OC or both), have been included in the review. A key aspect of this review is that it sheds light on the potential and limitations of AI in oral cancer screening, supported by suitable evidence. This review will also lay the groundwork for future research and implementation strategies, emphasising the transformative impact of AI in enhancing early diagnosis, reducing mortality, and addressing critical healthcare gaps, especially in underserved regions.

The findings of our study will provide evidence to support informed decision-making in the implementation of AI-based screening for oral cancer. By synthesising current evidence, this review supports decision-makers in selecting the most appropriate and effective AI technologies to ensure the efficiency and accuracy of screening programmes. Notably, implementing AI-driven oral cancer screening is promising for LMICs, which can significantly enhance access to early detection services for underserved and high-risk populations. By demonstrating the viability of AI-based screening tools within public health systems, our research contributes to advancing health equity and represents a meaningful step toward the broader goal of achieving universal health coverage.

In this review, most studies used imaging modality as the input, with a very few studies using data based on saliva samples. In the studies where smartphones were used for imaging the oral cavity, techniques, such as grids and centering were applied to ensure good image capture. Issues such as image blur and incorrect angle of capture may hinder AI performance, and image quality filters may be considered to enhance the quality of captured images. Most studies employed data augmentations, such as scaling, flipping, rotation, translation, random cropping, mosaic, mixup, and saturation, to increase the number of datapoints for training and to add variability to the training data. Studies also employed various image pre-processing techniques, such as cropping and filtering, to enhance diagnostic features in the images. Due to the lack of large datasets of oral images, studies developing DL techniques for oral cancer screening leveraged transfer learning principles, such as utilising weights on large datasets such as ImageNet as the initial point for training. Depending on the application, different adjudication mechanisms were used to determine the ground truth. Some studies also validated AI performance against humans on a subset of the validation dataset. Further, a few studies assessed human-in-the-loop AI systems for potential performance improvements.

Evidence suggests that AI can improve screening outcomes and reduce overall mortality and morbidity. However, for any further studies in this domain, we note that the following points should be considered:

- Many studies develop AI algorithms on considerably small datasets, which hinders the ability of the model to generalise to real-world scenarios. Furthermore, such datasets might not represent the prevalence of various lesions in the population. A well-sampled, multi-centre, population-representative dataset should be used to develop robust AI models.
- AI models are as good as the data used to train them. Apart from a well-sampled and representative dataset, proper data annotations (based on the application, screening or diagnosis) are crucial in developing efficient models. Data obtained from internet searches and publicly available datasets of questionable credibility require double-checking before being used for AI model development.

- There is a lack of validation of the developed AI models on external datasets. Validation of AI models on well-sampled population-scale data would enhance the reliability of the tools developed. Such validation would ensure that the AI technology is both accurate and appropriately tailored to the population in which it is implemented, accounting for the population-specific characteristics.
- In many instances, there is a significant domain shift in the datasets used to develop AI models and the data from real-world scenarios. The variability can come from factors such as lighting conditions and the device used to capture images, among others. Studies might consider assessing models' performance in various field conditions to enhance practical applicability.
- Very few articles included interpretability analysis for the developed models. Interpretability studies lend greater relevance to the use of AI tools in clinical practice.
- Investigating the potential reasons for misclassification by AI, such as a stratified analysis, would help in understanding the factors that affect the AI model's performance, aiding in efficient and improved training of AI models.
- Validating the performance of AI against human readers on external validation data helps assess the performance and biases that might guide clinical adoption of AI tools.

Limitations

This study has certain limitations that should be acknowledged. Firstly, including only English-language articles may have introduced a language bias, potentially excluding relevant evidence published in other languages. Second, some studies had to be excluded from the meta-analysis due to insufficient or incomplete information, which might have influenced the robustness and generalisability of the overall findings. Secondly, this review is limited by evidence of publication bias, as indicated by significant asymmetry in Deeks' funnel plot. This suggests that smaller, less favourable studies may be underrepresented, potentially leading to an overestimation of the pooled diagnostic performance. To mitigate this, we interpreted the pooled estimates cautiously and emphasised the consistency across larger, higher-quality studies. Future research should prioritise prospective, well-designed diagnostic accuracy studies and ensure that positive and negative findings are reported to reduce publication bias. Lastly, the majority of AI technologies identified in the included studies were image-based, reflecting a limited scope of available applications and potentially overlooking other promising modalities of AI in oral cancer screening. When interpreting the results and planning future research, these constraints should be considered to ensure a more comprehensive understanding of the field.

CONCLUSION

This study demonstrates that AI-enabled screening tools achieve high diagnostic accuracy in the early detection of oral cancer. These findings underscore the promise of AI as a complementary approach to conventional screening, with relevance for resource-limited settings. Future research should emphasise the use of large, representative datasets, rigorous external validation, and interpretability analyses to facilitate safe, equitable, and clinically robust implementation.

Abbreviations

AI: artificial intelligence

OSCC: oral squamous cell carcinoma

PRISMA: Preferred Reporting Standards for Systematic Reviews and Meta-Analyses

QUADAS: Quality Assessment of Diagnostic Accuracy Studies

ML: machine learning

DL: deep learning

CNN: convoluted neural network

WLI: white light-based imaging

AFI: autofluorescence imaging

PPV: positive predictive value

NPV: negative predictive value

TP: true positive

TN: true negative

FP: false positive

FN: false negative

Declarations

Ethics approval and consent to participate

Ethical approval was not required for this study as it is a systematic review of previously published literature and did not involve primary data collection from human participants.

Consent for publication

Not applicable.

Availability of data and materials

Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

Conceptualisation: DS, SG; Methodology: DS, SG; Data collection and analysis: DS, TS, AA; Supervision: DP; Writing — original draft: DS, TS, AA; Writing — review and editing: SG, DP.

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Table

Table 1. Characteristics of included studies

Sl. No.	Study	Location	Sample size	Population	Intervention	Comparator	AI model/tool (Classification/detection task)
1	Song 2024	India	2434 oral images, with 1243 'non-suspicious' (normal and benign) images, and 1191 'suspicious' (oral potentially malignant lesion and malignant lesions) images	Setting: Outpatient clinics; participants > 18 years old with a history of smoking/chewing tobacco or with any oral lesion	Vision Transformer Swin Transformer	Direct telediagnosis by specialists; oral images were annotated by oral oncology specialists	ViT (Suspicious vs. Non-suspicious) Swin Transform (Suspicious vs. Non-suspicious)
2	Goswami 2024	India	630 oral images sourced from a public dataset, (1200 lesion areas pulled, with 600 malignant and 600 normal). Another in-house dataset sourced through Google and Mozilla search engines (total 200 images with 100 images of leukoplakia and 50 each of erythro-leukoplakia and erythroplakia)	Dataset containing 630 oral images collected from internet sources and different hospitals and colleges	The method involves exploring five distinct colour spaces and extracting colour and texture features, which are then classified using a ML technique called Light Gradient Boosting Machine (LightGBM).	Consultation with oral specialists	LightGBM for binary classification (Benign vs. Malignant) LightGBM for multi-class classification (Leukoplakia vs Erythro-leukoplakia vs. Erythroplakia)
3	Mira 2024	Egypt	1448 images (760 normal, 251 aphthous ulcer, 231 low-risk OPMD, 141 high-risk OPMD, and 65 oral cancer images) sourced from 787 patients; test set consists of 455 images (228 normal, 76 aphthous ulcer, 69 low-risk OPMD, 52 high-risk OPMD, and 30 oral cancer images)	Adult healthy subjects classified as either (ASA-I) or (ASA-II) and clinically diagnosed with aphthous ulcer, OPMD, or oral cancer	The centred rule method of image-collection along with resampling method to mitigate the influence of image variability from handheld smartphone cameras. HRNet-W18, a CNN model is trained for five-class disease classification of these images.	Annotation of photographs by three oral medicine specialists based on the presentation of symptoms and clinical records or histopathology reports	Hrnet W-18 (Normal + Ulcer vs. OPMD + Cancer)
4	Banavar 2023	Australia, India	230 samples (20 OSCC, 76 OPSCC, 14 OPMD, 120 cancer free)	Individuals ≥ 50 years old or anyone with a history of tobacco use. The validation cohort of 230 included 101 samples from secondary care centres, and a combination of 129 clinically adjudicated patients with OPMD and cancer free patients from the Viome company customer database in the US.	Saliva metatranscriptomic detection test, CancerDetect for Oral & Throat cancer™ (CDOT) (ML model used: L2-regularised logistic regression)	OSCC or OPSCC: Clinical diagnosis of after biopsy and histopathology; OPMD and no cancer: Clinical adjudication by a primary physician. Each photo is classified as either (1) normal mucosa with no lesion, (2) clearly benign lesion, (3) lesion mimicking cancer, or (4) OSCC tumour. Classes 3 and 4 represent the suspicious cases that are confirmed through biopsy. Class 1 is determined solely by the physician, while class 2 may sometimes be confirmed through biopsy.	CDOT (OSCC + OPSCC vs. OPMD + Cancer free)

5	Rabinovici-Cohen 2024	Israel	Dataset of 2398 clinical photographic images from 1470 patients with 1382 photos from the selected 586 patients in Tzafon Medical Center in Israel and 1016 photos from 884 patients selected from online academic sources; in both cancer detection and suspicious detection tasks, the development set consisted of 1312 patients (2032 images) and the holdout test set consisted of 158 patients (366 images)	Cohort of 132852 patients who sought services at the oral and maxillofacial department	Two primary methods: CNN method employed using the ResNet18 architecture and Optimized CNN (OCNN) constructed upon ResNet50 with CNNmeta and OCNNmeta incorporating a late fusion of the metadata. An ensemble pf models based on the mean value of all the available scores per patient.	Each photo is classified as either (1) normal mucosa with no lesion, (2) a clearly benign lesion, (3) a lesion mimicking cancer, or (4) an OSCC tumour. Classes 3 and 4 represent the suspicious cases that are confirmed through biopsy. Class 1 is determined solely by the physician, while class 2 may sometimes be confirmed through biopsy.	CNNmeta for cancer detection (OSCC vs. other) OCNN for cancer detection (OSCC vs. others) Ensemble for cancer detection (OSCC vs. other) CNNmeta for suspicious detection (OSCC vs. other) OCNN for suspicious detection (OSCC vs. other) Ensemble for suspicious detection (OSCC vs. other)
6	Talwar 2023	India	Dataset of 1120 suspicious and 1058 non-suspicious oral cavity photographic images with 422 (234 suspicious and 178 non-suspicious) images used for testing	The patients included in the study were above 18 years of age. The intraoral images were collected by front-line healthcare workers.	DenseNet201 and Swin Transformer (base)	Three oral medicine specialists provided an image-level annotation and classified them into suspicious and non-suspicious images.	Densenet 201 (Suspicious vs. Non-suspicious) Swin Base (Suspicious vs. Non-suspicious)
7	Ye 2024	China	1419 oral photographs (262 OLP, 328 OLK, 799 OSCC, and 30 normal oral mucosa images) used for model development, internal evaluation, and external evaluation; the external test set included 30 images each of normal mucosa, OLP, OLK, and OSCC	1419 oral photos from smartphones and digital cameras were collected from a hospital.	YOLOX-based CNN model for efficiently detecting OLP, OLK, and OSCC in patient images integrated into a mobile application called "Oral Tec".	Lesion areas in 1419 photos classified by three oral specialists and two ML engineers based on biopsy confirmation and typical clinical manifestations	Macro-average (calculated across all categories: OLP, OLK, OSCC, Normal) performance values of the YOLOX model (evaluated on the external dataset)
8	Shree 2024	India	60 unstimulated whole saliva samples (from 30 histopathologically confirmed OSCC cases and 30 healthy controls), further divided into training set (80% of data) and test data set (20% of data)	60 patients divided into two groups- Group A included 30 histopathologically confirmed OSCC patients and Group B included 30 healthy controls.	ATR-Fourier transform infrared (FTIR) spectroscopy combined with SVM based ML techniques	Histopathological confirmation	SVM classifier (Cancer vs. Normal)
9	Vinayahalingam 2024	Germany	4161 clinical oral photographs with 3337 used for training and validation and 824 used for testing	4161 photographs of healthy mucosa, leukoplakia, OLP, and OSCC	Mask R-CNN model used along with the small variant of the Swin Transformer as backbone.	Each image labelled and pixel-wise segmented by different clinicians independently; subsequently reviewed and revised by three clinicians	Mask R-CNN with Swin transformer (OLP + Leukoplakia + OSCC vs. No disorder)
10	Warin 2021a	Thailand	600 (300 OPMD and 300 normal oral mucosa images) of which 420 (210 OPMD and 210 normal oral mucosa images) used for training, 60 (30 OPMD and 30 normal oral mucosa images) for validation, and 120 (60 OPMD and 60 normal oral mucosa images) for testing	600 images collected and divided into OPMD and normal oral mucosa groups of 300 images each	Two CNN-based classification algorithms DenseNet-121 and ResNet-50. Two CNN-based object detection algorithms Faster R-CNN and YOLOv4 were used to locate the OPMD lesions in the oral photographic images.	All OPMD images were biopsy-proven; OPMD images were annotated by three oral and maxillofacial surgeons with the largest area of intersection between all the surgeons' annotations as the ground truth.	DenseNet-121 (OPMD vs. Normal) ResNet-50 (OPMD vs. Normal) Faster R-CNN (for detecting lesion area in OPMD images) YOLOv4 (for detecting lesion area in OPMD images)

							area in OPMD images)
11	Birur 2022	India	30248 images from 4728 subjects; test sample size for MobileNet: 1416 image pairs, VGG19: 1167 image pairs	Individuals ≥ 18 years old with a history of tobacco smoking and/or chewing or with any oral lesion; participants were recruited from the tertiary cancer centre, dental hospitals, low resource settings, like primary health centres, and community camps, by Front-Line-Health-Workers	Phone-integrated CNN based on MobileNet, deployed with FHWs for low-resource settings and cloud-based CNN, VGG19-BDL: VGG19 Convolution Neural Network with Bayesian prediction a more accurate cloud-based classification.	Remote diagnosis by specialist	MobileNET (in mobile phone) \ remote diagnosis by specialist (Suspicious vs. Non-suspicious VGG19 (in cloud BDL uncertainty 0.15) vs remote diagnosis by specialist (Suspicious vs. Non-suspicious
12	Yang 2023	USA	2198 (1099 pairs of AFI and pWLI) images of healthy oral mucosa and oral lesions suspicious for OPML or OC collected from 318 subjects (247 self-identified non-smokers and 71 as smokers); the smoking group had 119 'no increased risk' and 126 'increased risk' image pairs; the non-smoking group had 667 'no increased risk' and 187 'increased risk' image pairs	Individuals with oral lesions suspicious of OPML or OC along with individuals with no visible oral lesions were recruited. Tobacco smokers were those who self-reported consuming 10 or more cigarettes per day and 70 cigarettes per week over the past year or longer. Tobacco non-smokers were those who had not smoked any tobacco within the past year or more.	The prototype oral cancer(OC) risk screening platform consists of a commercial Moto G5 Android smartphone, an intraoral imaging probe, a light-emitting diode driver, a rechargeable lithium battery, and the prototype mobile application (screening algorithm developed based on VGG-19).	Full standard of care OC screening performed by an oral medicine specialist with the outcome recorded as either "no increased risk" or "increased risk" for each study participant at all the study sites	VGG-19 (increased risk vs. no risk in Tobacco Non-Smoker group) VGG-19 (increased risk vs. no risk in Tobacco-Smoker group)
13	Warin 2022	Thailand	Dataset of 980 images (365 OSCC, 315 OPMD, 300 non-pathological images)	980 oral photographic images	CNN-based image classification networks, DenseNet-169, ResNet-101, SqueezeNet and Swin-S, for multi-class image classification(OSCC vs. OPMDs vs. non-pathological) of oral photographic images. CNN-based object detection algorithms, Faster R-CNN, YOLOv5, RetinaNet and CenterNet2, for detection of OSCC and OPMDs lesions in oral photographic images.	All of the OSCC and OPMDs images were biopsy-proven confirmed by oral pathologists as the gold standard for diagnosis. The lesion boundaries of the OSCC and OPMDs images were annotated by three oral and maxillofacial surgeons with ground truth being the largest area of intersection between all of the surgeons' annotations.	DenseNet-169 (OSCC classification) ResNet-101 (OSCC classification) SqueezeNet (OSCC classification) Swin-S (OSCC classification) DenseNet-169 (OPMDs classification) ResNet-101 (OPMDs classification) SqueezeNet (OPMDs classification) Swin-S (OPMDs classification) Faster R-CNN (OSCC detection) YOLOv5 (OSCC detection) RetinaNet (OSCC detection) CenterNet2 (OSCC detection) Faster R-CNN (OPMDs classification)

							detection) YOLOv5 (OPMD detection) RetinaNet (OPN detection) CenterNet2 (OPMDs detection)
14	Song 2018	India	Dual-mode (AFI and WLI) image pairs from 190 patients, 170 (84 suspicious and 86 normal) good-quality image pairs with 20 image pairs of both categories used for validation	Outpatient clinic patients clinically diagnosed with OPML and malignant lesions	Dual model CNN classification method based on VGG-CNN-M model	Clinical diagnosis; images labelled by oral oncology specialists	VGG-CNN-M on the dual-modal images (suspicious vs. normal)
15	Warin 2021b	Thailand	700 (350 OSCC and 350 normal) clinical oral photographs with 490 images used for training, 70 for validation, and 140 for testing	700 clinical oral photographs retrospectively collected from oral and maxillofacial surgery center; 350 images of OSCC and 350 of normal oral mucosae	The classification and detection models created using DenseNet121 and Faster R-CNN, respectively	All OSCC images were biopsy-proven; OSCC images annotated by three oral and maxillofacial surgeons with the largest area of intersection between all of the surgeon's annotations as the ground truth	DenseNet121 (OSCC vs. Norm) Faster R-CNN (f detecting lesion area in OSCC images)
16	Song 2021	India	7970 pairs (AFI and WLI) of intraoral images collected from 5025 patients; 6211 pairs of images passed the quality test: 5329 images used for training and validation and 882 image pairs used for testing	5025 patients attending outpatient clinics	Dual-modality mobile oral screening device and Android smartphone application with trained MobileNet model implemented for image classification	Clinical diagnosis; image pairs labelled by oral oncology specialists as 'normal' (containing normal and benign mucosal lesion images) and 'suspicious' (containing OPML and malignant lesion images)	MobileNet implemented in smartphone application (suspicious vs. normal)
17	Nanditha 2021	India	332 oral lesion images (63 benign lesions and 269 pre-cancerous lesions), 80% is used for training and 20% of this data used for testing	Oral lesion images have been collected from different colleges and hospitals	The images have been captured using mobile phone cameras and intraoral cameras. Two DL models have been considered - residual network-50 (ResNet-50) and skip connected VGG. These two models have been ensemble with hybrid (frequency domain + spatial domain) pre-processing technique for better classification results	Images with labels were sourced from different colleges and hospitals	Ensemble of ResNet-50 and skip connected VGG with hybrid pre-processing technique (Malignant vs. Benign)
18	Fu 2020	China	5775 photographs (development dataset) used to develop the algorithm and 401 photographs (internal validation dataset) used for validation; an external validation dataset with 402 clinical photographs and a clinical validation dataset of 666 images used as additional validation datasets	44409 clinical oral photographs with all biopsy-proven OCSCC photographs and normal controls from 11 hospitals; an external validation dataset of 420 clinical photographs from six representative journals in the field of dentistry and oral and maxillofacial surgery, and a	An automated DL algorithm using cascaded CNNs (Single Shot MultiBox Detector with VGG16 as a backbone network for lesion detection followed by a DenseNet-based classifier) to detect OCSCC from photographic images	Biopsy-proven OCSCC photographs	Internal validation dataset (oral cancer lesions, Negative controls) External validation dataset (oral cancer lesions, Negative controls)

				clinical validation dataset from outpatient departments containing 1941 photographs of OCSCC, other diseases or disorders of oral mucosa, and normal oral mucosa			Clinical validation dataset (oral cancer lesions vs. Negative controls)
19	Lin 2021	China	688 oral lesion images (251 aphthous ulcer, 231 low-risk OPMD, 141 high-risk OPMD, and 65 cancer images) and 760 normal mucosa images; the test set (455 cases) had 228 normal mucosa images, 76 aphthous ulcers, 69 low-risk OPMD, 52 high-risk OPMD, and 30 cancer images	Outpatient and inpatients >18 years of age, including healthy people and those clinically diagnosed with aphthous ulcers, OPMD, or oral cancer	Handheld smartphone camera and CNNs technology: high-resolution representation learning network (HRNet) for oral disease classification	Each image annotated by two senior oral medicine experts with reference to the symptoms, clinical records or histopathology reports and reviewed by a third oral medicine specialist	HRNet-W18 (Low risk OPMD+High risk OPMD+oral cancer vs. Normal+Aphthous Ulcer)
20	Kouketsu 2024	Japan	1043 images of lesions from 424 patients; evaluation data had 66 images of oral cancer, 49 of leukoplakia, 405 of other diseases not requiring closer examination	Study group: patients clinically diagnosis with oral mucosal disease, including oral cancer and leukoplakia of the tongue, and scheduled for surgery; control group: patients with other benign tongue pathologies	AI-based screening methodology using oral cavity images captured with a single-lens reflex camera. (Single Shot Multibox Detector was adopted as a DL method)	Histopathology results of the resected specimens	SSD: (OSCC vs. normal) SSD: (OSCC + leukoplakia vs. normal)
21	Jubair 2022	Jordan	716 (236 suspicious and 480 benign) clinical images from 543 patients; training, validation and test sets contained 566, 50, and 100 images respectively	Clinical images for various tongue lesions of patients	Lightweight deep CNN, EfficientNet-B0 used as a transfer learning model to develop a binary classifier of clinical images of the tongue into either benign or suspicious (cancer or OPMD)	Definitive diagnoses of the included images confirmed by reviewing patient records and pathology reports	Efficient-B0 (Suspicious vs. Benign)
22	Uthoff 2018	India	Data collected from 99 patients with 170 image pairs (WLI and AFI) divided into a normal class (N = 86) and a suspicious class (N = 84)	Subjects with clinically suspicious oral lesions, a history of previously treated OSCC with no current evidence of cancer recurrence at least 6 months after cessation of treatment, or the presence of recently diagnosed, untreated OSCC or pre-cancerous lesions	Smartphone-based oral cancer screening device to collect both AFI and WLI with both the whole cavity imaging module and the intraoral probe module. Further suspicious/not-suspicious classification by CNN (VGG-M) and remote specialist diagnosis	Each lesion site clinically diagnosed by an oral oncology specialist, with the clinical diagnosis serving as the gold standard (on-site specialist diagnosis)	VGG-M (Suspicious vs. Normal)
23	Shamim 2022	Saudi Arabia	Binary dataset with 200 (100 benign and 100 pre-cancer) annotated images randomly split into training (n = 160) and validation (n = 40) sets	Clinically annotated photographic images of tongue lesions	For classifying tongue lesions as benign or pre-cancerous, prediction models based on different DCNN architectures (AlexNet, GoogLeNet, VGG19, Inceptionv3, ResNet50 and SqueezeNet) were built with best performance seen in VGG19	Photographic images of the tongue manually annotated by a certified physician with more than 15 years of clinical experience	VGG19 (Pre-cancerous vs. Benign)
24	Sharma 2022	India	Dataset of 329 images with 106 normal images, 102	Retrospectively collected clinical	Pre-trained CNN architecture namely	Biopsy-proven OSCC, OPMLs,	VGG19 (Pre-malignant +

			OPMD images, and 121 OSCC images used; 85% of the data used for training the CNNs and 7.5% each used for validation and testing	oral photographs from patients with OSCC or OPMDs and comparison them with photographs of normal oral mucosa sourced from the archives of the Department of Oral Medicine and Radiology and the Department of Oral and Maxillofacial Surgery	MobileNet, VGG19, VGG16, InceptionV3, and ResNet50 were considered for experimentation using transfer learning	and normal mucosa images	Malignant vs. Normal) VGG16 (Pre-malignant + Malignant vs. Normal) MobileNet (Pre-malignant + Malignant vs. Normal) InceptionV3 (Pre-malignant + Malignant vs. Normal) ResNet50 (Pre-malignant + Malignant vs. Normal)
25	Dutta 2024	India	716 clinical images from 543 patients with 236 images of SCC or oral epithelial dysplasia and 480 images of various benign tongue lesions; test set consisted of 100 images (suspicious: 50, benign: 50) from 100 patients (suspicious: 50, benign: 50)	Clinical images of various tongue lesions	RBFN-SDC model	Definitive diagnoses of the included images confirmed by reviewing patients' medical histories and pathology results	RBFN-SDC (suspicious vs. benign)
26	Sampath 2024	India	131 images (87 cancer and 44 non-cancer); 70% of the images randomly selected for training the model, and 30% used for evaluating the model's performance	Images were sourced from various ENT Ear, hospitals	OralNet with Feature Fusion Deep Convolution Neural Network (ResNet-50) with Stochastic Gradient based Logistic Regression for classification	Expert classifications performed in collaboration with ENT doctors	OralNet (Cancer vs. Non-cancer)
27	Banavar 2021	Australia	433 saliva samples and metadata from 242 unique individuals	71 newly diagnosed treatment-naïve patients with OC or OPMD and 171 non-diseased individuals were categorised into high-risk population (age ≥ 50 years OR a current or past smoking) and average-risk population (general population across all backgrounds and histories)	Combining salivary microbial transcriptome (metatranscriptome) profiling using next-generation sequencing technology with ML (L2-regularised logistic regression)	OC: clinical assessments for performed using standard-of-care biopsies and histopathology evaluations. Normal healthy controls: state of health assessed using a survey questionnaire on recent history of habits or other medical conditions	ML classifier or Cohort A (high-risk OC + OPMD discovery cohort for OC+OPMD vs. normal classification) ML classifier or Cohort B (high-risk cross-validation cohort) for OC+OPMD vs. normal classification ML classifier or Cohort C (average risk OC only) for OC+OPMD vs. normal classification

OCSCC: oral cavity squamous cell carcinoma; OPSCC: oropharyngeal squamous cell carcinoma; SCC: squamous cell carcinoma

Figures

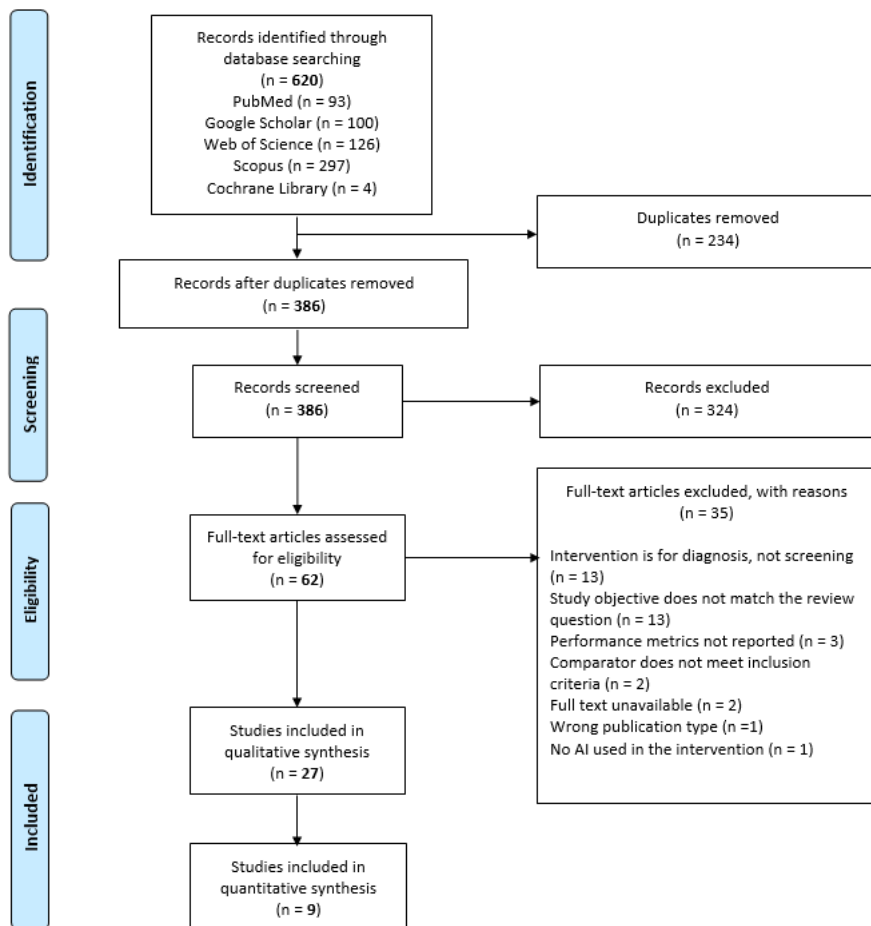


Figure 1
PRISMA flowchart

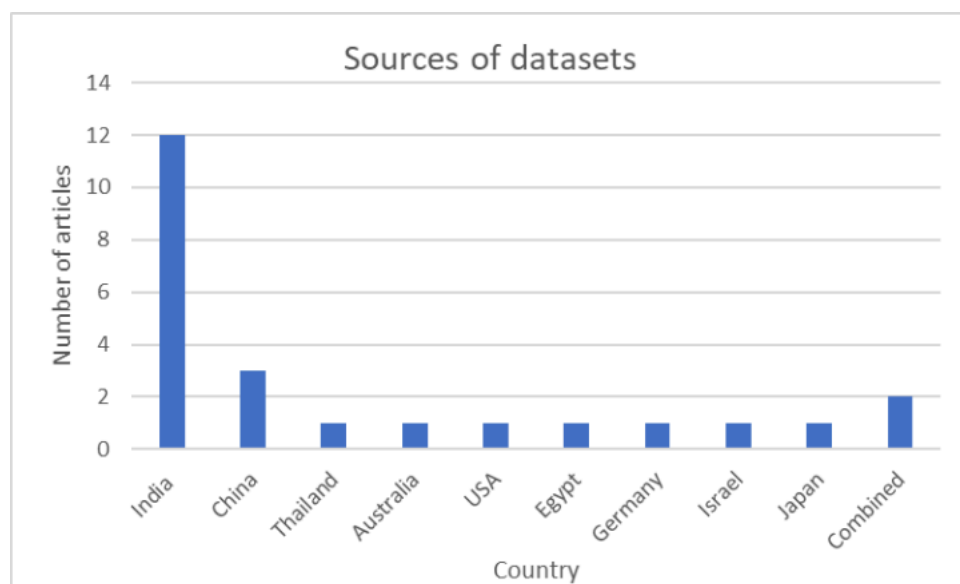


Figure 2
Sources of datasets used in the studies included in this review

A.

	Risk of bias domains				
	D1	D2	D3	D4	Overall
Song 2024	+	+	+	+	+
Goswami 2013	?	+	+	+	+
Mira 2024	+	+	+	+	+
Banavar 2023	+	+	+	+	+
Rabinovici-Cohen 2024	+	+	+	+	+
Talwar 2023	?	+	+	+	+
Ye 2024	+	+	+	+	+
Shree 2024	+	+	+	+	+
Vinayahalingam 2024	+	+	+	+	+
Warin 2021a	+	+	+	+	+
Birur 2022	+	+	+	+	+
Yang 2023	+	+	+	+	+
Warin 2022	+	+	+	+	+
Song 2018	+	+	+	+	+
Warin 2021b	+	+	+	+	+
Song 2021	+	+	+	+	+
Nandiha 2021	?	+	+	+	+
Fu 2020	+	+	+	+	+
Lin 2021	+	+	+	+	+
Kouketsu 2024	+	+	+	+	+
Jubair 2022	+	+	+	+	+
Uthoff 2018	+	+	+	+	+
Shamim 2022	?	+	+	+	+
Sharma 2022	+	+	+	+	+
Dutta 2024	+	+	+	+	+
Sampath 2024	?	+	+	+	+
Banavar 2021	+	+	+	+	+

Domains:
D1: Patient selection.
D2: Index test.
D3: Reference standard.
D4: Flow & timing.

Judgement
+ Low
? No information

B.

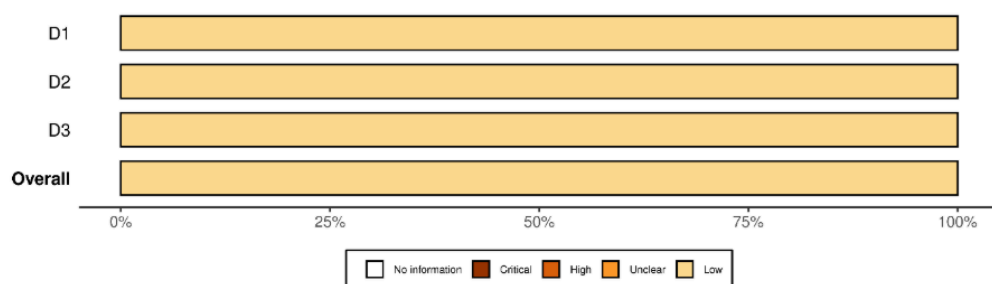


Figure 3

QUADAS-2 assessment: A. Risk of bias; B. Applicability concerns

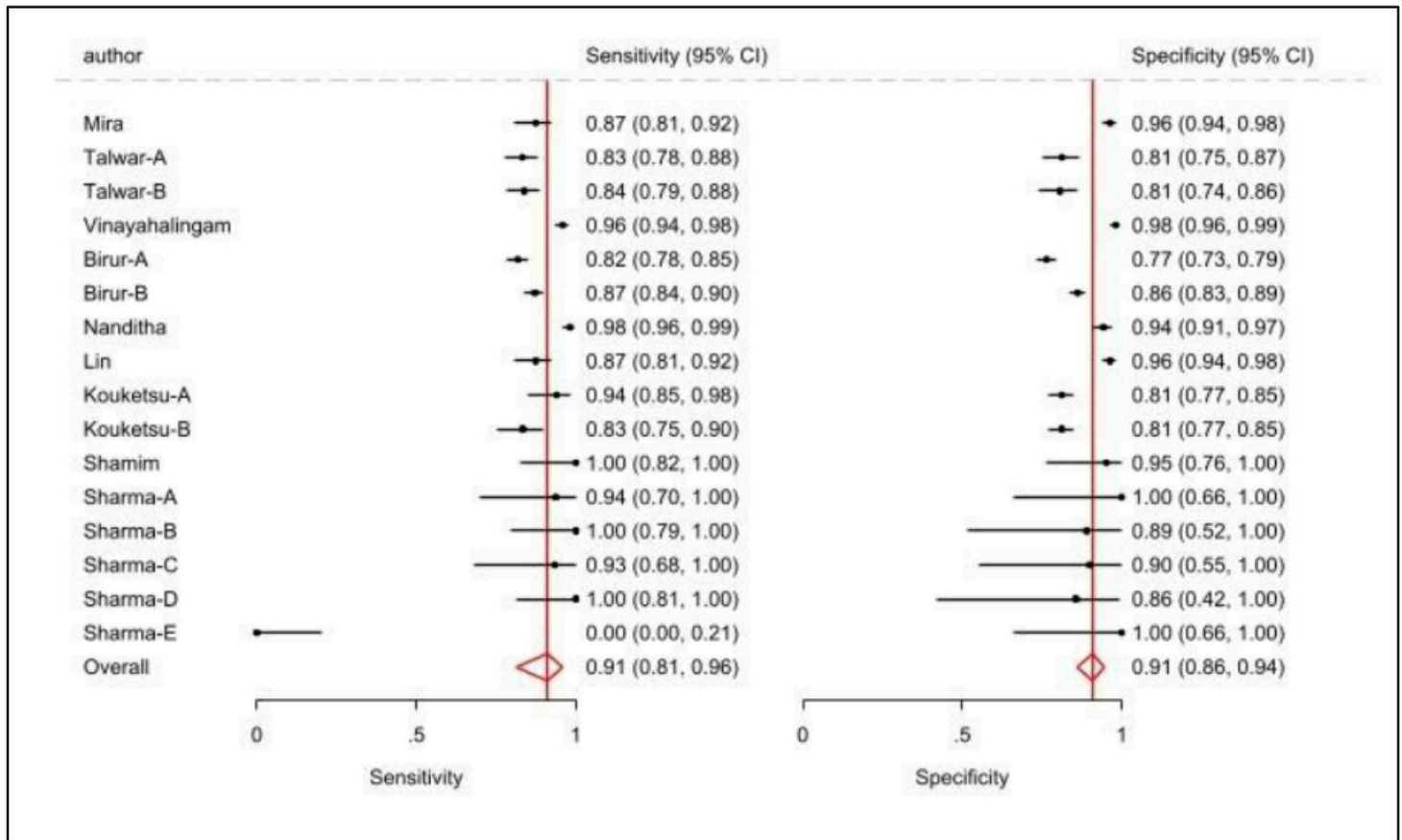


Figure 4

Forest plot showing the pooled sensitivity and pooled specificity

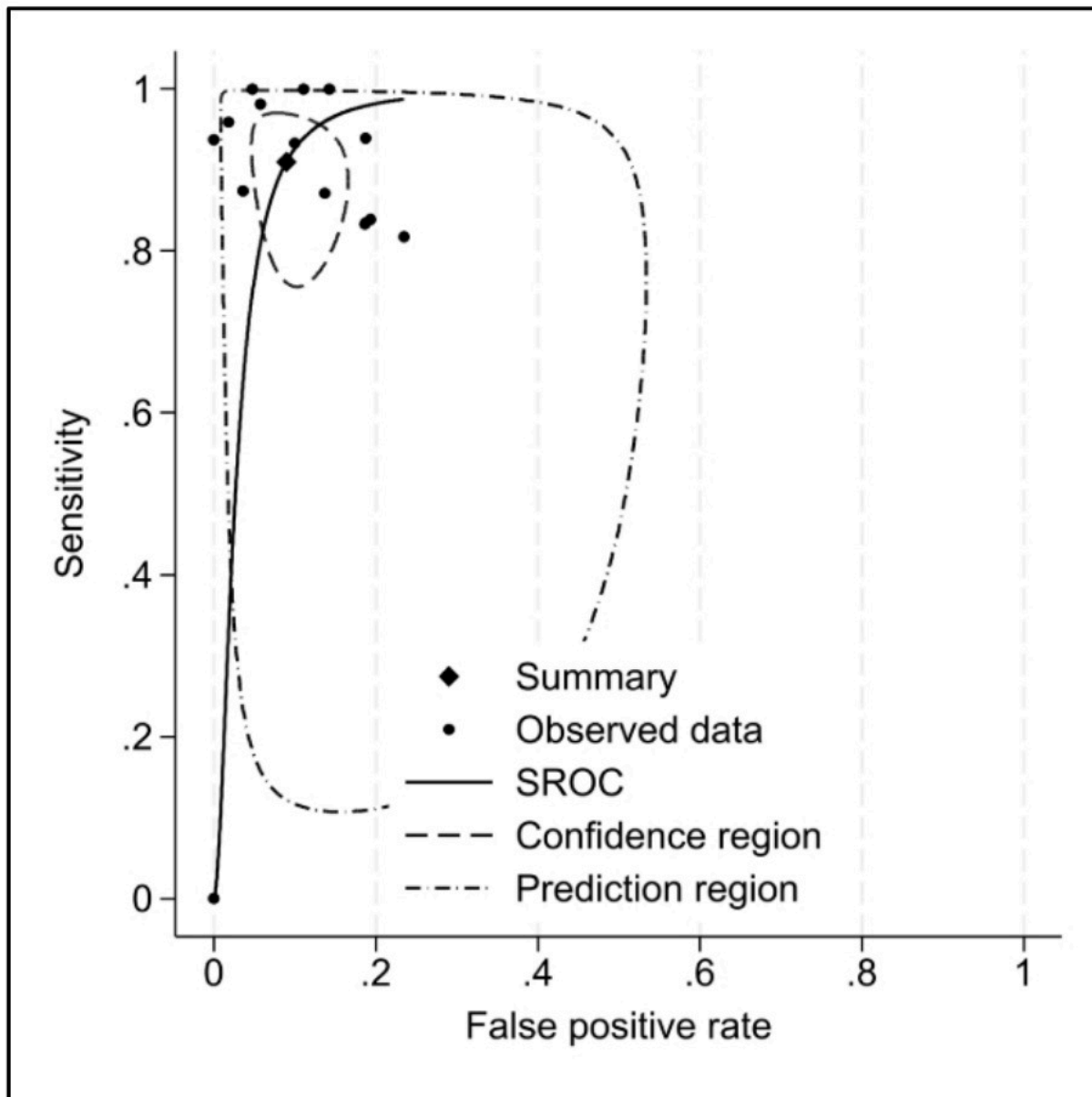


Figure 5

Summary receiver operating characteristic curve

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [SupplementaryInformationDiagnosticAccuracyofAIEnabledScreeningforOralCancerSRMA.pdf](#)